Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Interaction pair

Meaning in my words How would you explain interaction pair to a friend?	**How can you tell?** Earth pulls an apple down. What is the partner force, and which object feels it?
Example or sketch How would you label the pair when a hand pushes a cart?	**Non-example and why** Are the table's push and Earth's pull on one book a third-law pair?

Equal and opposite

Meaning in my words What does "equal and opposite" tell you about a pair of interaction forces?	**How can you tell?** One skater pushes another with 30 N east. What force does the first skater feel from the second?
Example or sketch Draw the arrows for two carts whose bumpers press together.	**Non-example and why** Do equal arrows guarantee equal accelerations for a heavy and a light skater?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Force receiver

Meaning in my words How would you explain force receiver to a friend?	**How can you tell?** In “the wall pushes on my hand,” where does that force arrow belong?
Example or sketch When my hand pushes on the wall, which object receives that other force?	**Non-example and why** Can I draw both contact forces of that pair on the hand?

Mass and acceleration

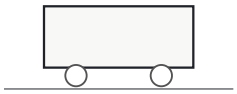
Meaning in my words How does mass affect acceleration for the same net force?	**How can you tell?** Two skaters push off with equal net force magnitudes. Which skater has greater acceleration?
Example or sketch A 10 N net force acts separately on 2 kg and 5 kg carts. Compare their accelerations.	**Non-example and why** The lighter cart accelerates more. Does that prove it felt the larger interaction force?

Doodle the idea

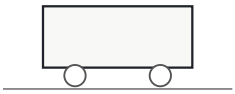
SECTION B / DRAW WITH YOUR TEACHER Name _____

Label both receivers first. Draw equal force arrows on different objects, pointing toward one another for gravity.

Receiver A



Receiver B



My marks mean

Two pulls. Two receivers.

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SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Place an interaction pair on its two receivers.

CHOOSE ONE JOB TOGETHER

SIMULATION ENGINEER ☐

Put each gravitational arrow on the body that receives it. Send the animator two corrected object panels.

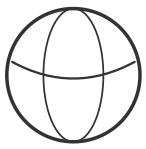
SCIENCE GRAPHICS PRODUCER ☐

Write a split-screen caption naming who pulls whom. Explain why equal force does not require equal acceleration.

Point to a clue. Keep your first marks; add the update using the color key.

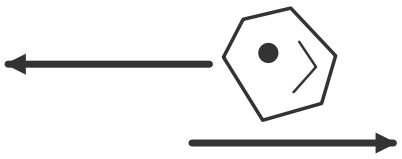
Earth pulls on XJ. A public animation shows both gravitational arrows on XJ, with the caption “they cancel.”

EARTH PANEL



Earth

XJ PANEL



PROPOSAL: “They cancel.”

01 OUR FIRST CALL

Do the two arrows cancel on XJ? Explain using your corrected panels.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Which arrow belongs in the updated shot? Name its source and receiver.

Switch speakers. Which object receives each force?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks